

Report
Of
**AI-ASSISTED CISCO NETWORK TROUBLESHOOTING AND
DIAGNOSTIC SYSTEM**

NETSAGE AI

Submitted
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B.Tech CSE



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Batch: 2023-27

CANDIDATE'S DECLARATION

I, UTKARSH KUMAR do hereby declare that the internship report titled “**AI-Assisted Cisco Network Troubleshooting and Diagnostic System — NetSage AI**” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

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I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

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INTERNSHIP COMPLETION CERTIFICATE

Cisco Networking Academy	 IILM UNIVERSITY
This certificate is awarded to Utkarsh Kumar for successfully completing Introduction to Modern AI offered by IILM University through the Cisco Networking Academy program.	
 Swati Vashisht Instructor IILM University	12 Jun 2026 Completion Date
Cert ID: 81abf3e8-123e-4c43-b2f7-83aa2bddb43f	

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4. PROJECT DESCRIPTION

4.1 Introduction

Computer networks are essential for communication between computers, servers, applications, and users. In Cisco networking laboratories, a small configuration error in a VLAN, gateway, routing table, DHCP service, DNS configuration, ACL, NAT rule, interface, or wireless setup can cause a connectivity problem. Students and junior engineers may know individual Cisco commands but still find it difficult to connect a symptom with the actual root cause.

NetSage AI was developed to make this troubleshooting process more structured and easier to follow. The system reads a network symptom and supplied show-command or host-command evidence, identifies a likely cause using a rule-based diagnostic engine, reports the relevant OSI layer, recommends a verification command, and proposes safe fix steps. The diagnosis remains advisory and is not allowed to directly apply a configuration change.

The supplied implementation contains exactly 30 structured troubleshooting cases, a Python diagnostic engine, a deterministic checker, an interactive HTML/CSS/JavaScript dashboard, a structured diagnosis prompt, and a Responsible AI review workflow. The project therefore combines networking knowledge, Python programming, structured data, interface development, evaluation, and human oversight.

4.2 Organization Profile

The supplied project materials identify Cisco Networking Academy as the training platform. The work described in this report was completed as an independent internship in Artificial Intelligence and network troubleshooting and is separate from the academic curriculum of IILM University. No separate corporate internship profile or company certificate was included in the supplied files.

Cisco-style networking concepts form the practical foundation of the project. The scenarios are designed around Packet Tracer/lab-style evidence such as show-command outputs, host configuration information, VLAN status, routing information, ACL entries, NAT translations, DHCP information, and wireless notes. The project uses this evidence in a controlled dataset instead of connecting directly to live production devices.

The internship work was completed by Bhola Kumar. The project covered dataset preparation, diagnostic logic, deterministic validation, structured prompt design, dashboard development, Responsible AI review, testing, debugging, and final documentation.

4.3 Problem Statement

The project addresses the difficulty of diagnosing Cisco-style laboratory network faults from symptoms and command evidence. A learner may observe that a PC cannot reach a server, a VLAN does not work across a trunk, DHCP clients receive incorrect addresses, DNS resolution fails, or an ACL blocks application traffic. The challenge is to identify a likely root cause and the next verification step without inventing evidence or applying an unsafe configuration change.

The supplied specification therefore requires an AI-assisted troubleshooter that accepts symptoms, Packet Tracer notes, and show-command outputs; recommends a likely fault and OSI layer; identifies the evidence supporting the diagnosis; suggests a next diagnostic command; and provides evidence-backed fix steps. Every diagnosis must be reviewed by a human before a configuration change is accepted.

The project requirements were implemented using at least 30 cases, a structured prompt library, a deterministic Python checker, an interactive dashboard, and a Responsible AI review log. The final dataset contains exactly 30 cases and the review log contains 10 reviewed cases, including five accepted and five edited diagnoses.

4.4 Project Objectives

- Develop an AI-assisted rule-based diagnostic engine for common Cisco network faults.
- Prepare a structured set of 30 troubleshooting cases containing symptoms, evidence, expected faults, OSI layers, concepts, and severity.
- Create a structured diagnosis format containing root cause, confidence, evidence, next command, OSI layer, fix steps, and human-review status.
- Develop a deterministic Python checker that independently validates common configuration clues.
- Build an interactive dashboard for case metrics, issue types, severity, diagnosis details, and Responsible AI information.
- Require human review before any suggested configuration change is accepted.
- Record accepted and corrected responses so that diagnosis quality can be inspected and audited.
- Evaluate the implemented rule-based diagnosis against the known expected faults in the supplied dataset.

4.5 Scope of the Project

The scope is limited to Cisco Packet Tracer and Cisco-style networking laboratory troubleshooting. The system focuses on structured evidence rather than live device access. It is intended as an academic learning and diagnostic-support tool rather than an autonomous network management platform.

Functional Scope

- VLAN and trunk configuration problems.
- Default gateway, subnet mask, and IP addressing issues.
- DHCP and DHCP relay problems.
- DNS server and stale-record problems.
- Routing, static route, OSPF, and default-route problems.
- ACL and application-access problems.
- NAT/PAT translation and interface issues.
- Wireless VLAN and guest-isolation scenarios.
- Interface state, speed/duplex, SVI, and related Layer 1–3 faults.
- Responsible AI review through accepted and edited diagnosis records.

Out of Scope

- Direct connection to live production routers or switches.
- Automatic configuration deployment to network devices.
- Automatic learning of new rules from live traffic.
- A claim of real-world diagnostic accuracy beyond the supplied structured dataset.

4.6 Technologies and Tools Used

Technology / Tool	Use	Purpose
Python	AI diagnostic engine and deterministic checker	Rule-based diagnosis, evidence matching, validation
CSV	Case dataset and Responsible AI review log	Structured storage of cases and human-review records
HTML/CSS/JavaScript	Interactive dashboard	Case exploration, diagnosis display and review information
Cisco Packet Tracer	Laboratory environment	Cisco-style network scenarios and command evidence
Markdown	Structured prompt library	Prompt instructions, schema and worked examples

Table 4.1: Technologies and tools used

The Python engine loads the CSV dataset and applies ordered diagnostic rules to the symptom and evidence. The rules deliberately place more specific clues before generic fallbacks. The checker provides an independent validation layer for common findings. The dashboard is generated from the same case and review data and exposes the diagnosis and human-review workflow in separate sections.

The structured prompt library defines the expected input, the required JSON schema, safety constraints, and examples. It explicitly instructs the diagnostic process to use only supplied evidence, avoid guessing when evidence is insufficient, recommend a verification command, and mark human review as required.

4.7 System Architecture

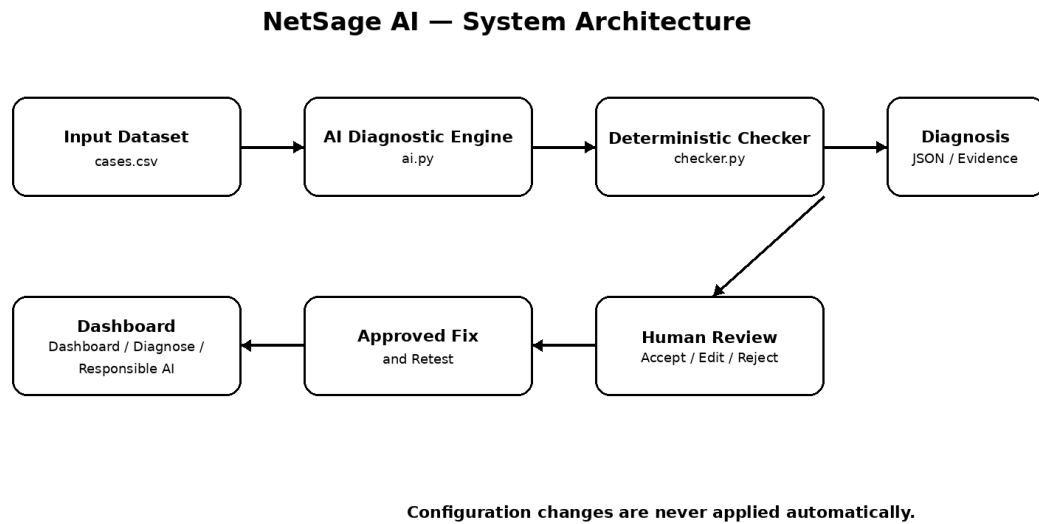


Figure 4.1: NetSage AI system architecture

The architecture begins with the structured cases.csv dataset, which stores the symptoms, evidence, expected fault, OSI layer, networking concept, and severity for each case. The ai.py module reads this information and applies ordered evidence-to-diagnosis rules. The output is a structured diagnosis containing root cause, confidence, evidence, next command, OSI layer, fix steps, and the mandatory human-review status.

The checker.py module operates as a deterministic validation layer. It searches for common clues such as an administratively down interface, missing routes, gateway mismatch, incorrect subnet mask, missing VLAN, native VLAN mismatch, DHCP relay issues, duplicate IP evidence, ACL deny rules, NAT mismatches, DNS problems, and speed/duplex mismatch. This independent check helps make the workflow more transparent.

The resulting information is presented through the dashboard. The dashboard contains three major areas: Dashboard for metrics and case distribution, Diagnose for case-level reasoning, and Responsible AI for human-review records. The implementation explicitly states that configuration changes are not applied automatically.

4.7.1 Dataset Coverage

Category	Cases	Severity represented
ACL	4	High/Medium
VLAN	3	High/Medium
Routing	3	High/Medium
Wireless	3	Critical/High/Medium
DHCP	2	High/Medium
DNS	2	Medium
NAT	2	High
Interface	2	High/Medium
Other network categories	9	Mixed

Table 4.2: Dataset coverage summary

Dataset-level distribution: 17 High, 12 Medium and 1 Critical cases. OSI distribution includes 15 Layer 3 cases and 5 Layer 2 cases, with additional Layer 4, Layer 7 and combined-layer scenarios.

4.8 Methodology

NetSage AI — Methodology Workflow

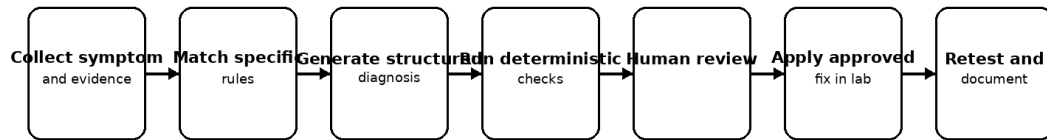


Figure 4.2: NetSage AI methodology workflow

The methodology follows a controlled evidence-to-decision workflow. First, the symptom and available network evidence are collected from the structured case. Second, the diagnostic engine checks the evidence against ordered rules, with specific patterns evaluated before general patterns. Third, the engine generates a structured diagnosis. Fourth, deterministic checks provide an independent validation layer. Fifth, a human reviewer evaluates the recommendation. Only after approval can a proposed configuration change be applied in the laboratory, followed by retesting and documentation.

4.8.1 Rule-Based Diagnosis

The diagnostic engine converts the symptom and evidence to lowercase and combines them for matching. Each rule contains a clue, a likely root cause, and a next diagnostic command. Examples include “administratively down” mapped to an interface-down diagnosis, “vlan 30 absent” mapped to a trunk allowed-list problem, “no ip helper-address” mapped to a DHCP relay problem, “wrong outside interface” mapped to a NAT/PAT problem, and “deny tcp” mapped to an ACL traffic-blocking diagnosis.

The implementation also contains special handling for OSPF because the generic fallback clue “absent” could otherwise produce a less specific diagnosis. The OSPF evidence is checked first and is mapped to the need to verify the OSPF network statement using the OSPF database command.

4.8.2 Structured Diagnosis and Safety

The structured prompt requires the diagnosis to contain `root_cause`, `confidence`, `OSI layer`, `evidence`, `next_command`, `fix_steps`, and `human_review`. It explicitly states that only supplied evidence may be used, that no command output or topology facts should be invented, and that insufficient or conflicting evidence should lead to a low-confidence response requesting more evidence.

The standard fix sequence in the implemented engine is: verify the finding with the suggested command, apply the approved configuration change, and retest the original symptom. The final field remains `human_review: required`. This design prevents the diagnostic assistant from being treated as an autonomous configuration agent.

4.9 Expected Outcomes and Achieved Results

4.9.1 Expected Outcomes

- A usable academic assistant for common Cisco lab troubleshooting cases.
- Evidence-backed root-cause suggestions instead of unsupported guesses.
- A consistent structured diagnosis format suitable for inspection and evaluation.
- Independent deterministic validation of common configuration clues.
- A dashboard that makes cases, severity, issue types, diagnosis details, and human review visible.
- A traceable Responsible AI process for accepting, editing, or rejecting recommendations.

4.9.2 Evaluation Results

The supplied checker output reports an evaluation match of 30/30 (100.0%) against the known expected faults in the structured dataset. This is a dataset-level rule-based evaluation and is not a claim of real-world diagnostic accuracy. All 30 cases received high confidence in the implemented rule-based output because each case was constructed around a matching evidence clue.

30 / 30 cases matched the expected fault at dataset level.

4.9.3 Dashboard

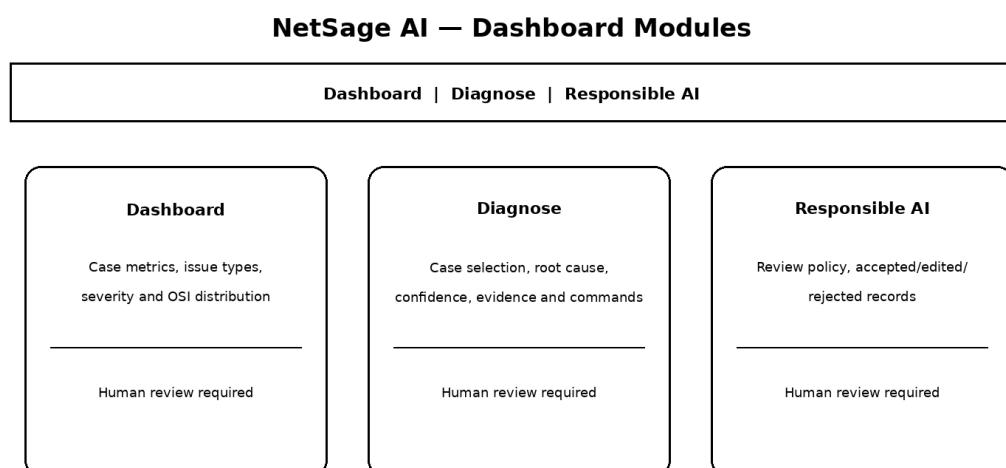


Figure 4.3: Dashboard module layout

The dashboard implementation presents three sections. The Dashboard summarizes 30 total cases, 18 high/critical cases, 17 issue types, five AI corrections, and 50% AI–human agreement based on the 10 reviewed records. The Diagnose section allows a user to select a case and inspect its root cause, confidence, OSI layer, evidence, next diagnostic command, and proposed fix steps. The Responsible AI section displays the review policy and the human-review log.

4.9.4 Test Case Summary

ID	Issue Type	Expected Fault	OSI	Severity
C001	VLAN	VLAN 20 is missing	Layer 2	High
C002	Gateway	Incorrect default gateway on PC	Layer 3	High
C003	DHCP	DHCP pool uses the wrong network	Layer 3	High
C004	DNS	PC points to the wrong DNS server	Layer 7	Medium
C005	Routing	Missing route to remote LAN	Layer 3	High
C006	ACL	ACL blocks required HTTP traffic	Layer 4	High
C007	NAT	PAT is applied to the wrong outside interface	Layer 3	High
C008	Wireless	Guest gateway has no usable default route	Layer 3	Medium
C009	Interface	Access interface is shut down	Layer 1/2	High
C010	Trunk	VLAN 30 is not allowed on the trunk	Layer 2	High
C011	Mask	Incorrect subnet mask	Layer 3	Medium
C012	Duplicate IP	Duplicate IP address detected	Layer 2/3	High
C013	SVI	Subinterface is not operational	Layer 3	High
C014	Native VLAN	Native VLAN mismatch	Layer 2	Medium
C015	OSPF	Remote network is not being advertised by OSPF	Layer 3	High
C016	Static Route	Static route to headquarters is missing	Layer 3	High
C017	ACL	ACL permits SSH from only one source subnet	Layer 4	Medium
C018	DHCP Relay	DHCP relay address is missing	Layer 3	High
C019	DNS	Stale DNS record	Layer 7	Medium
C020	NAT	Incorrect NAT translation rule	Layer 3	High
C021	Wireless	Guest isolation ACL is missing	Layer 3/4	Critical
C022	VLAN	Voice VLAN is missing	Layer 2	Medium
C023	Interface	Speed/duplex mismatch	Layer 1/2	Medium
C024	Routing	Default route points to stale next hop	Layer 3	High
C025	ACL	ACL permits ICMP but blocks HTTPS	Layer 4	Medium
C026	DHCP	DHCP pool is exhausted	Layer 3	Medium
C027	VLAN	Port is assigned to the wrong VLAN	Layer 2	Medium
C028	Routing	More specific static route is taking precedence	Layer 3	Medium
C029	Wireless	SSID is mapped to the wrong VLAN	Layer 2/3	High
C030	ACL	ACL denies management subnet access	Layer 3	High

Table 4.3: Thirty troubleshooting test cases

4.9.5 Responsible AI Review Results

Responsible AI is a central part of the project. The system does not automatically apply configuration changes. A human reviewer must approve, edit, or reject each diagnosis before a fix is accepted. The supplied review log contains 10 reviewed cases: five were accepted and five were edited. Therefore, the recorded acceptance/agreement rate is 50% for this review sample.

Case	Reviewer	Status	Human Review / Correction
C006	Network Reviewer 1	Edited	Confirmed the ACL blocks HTTP specifically; verify ACL direction before changing it
C015	Network Reviewer 1	Edited	Verify the OSPF network statement and area configuration before changing it
C020	Network Reviewer 2	Edited	Inspect the NAT rule and inside/outside interface definitions before modifying the translation
C025	Network Reviewer 2	Edited	Verify the ACL entry, interface direction, and HTTPS requirement before editing
C030	Network Reviewer 3	Edited	Confirm the ACL is applied to the management interface in the correct direction before editing the rule
C001	Network Reviewer 1	Accepted	No correction required
C002	Network Reviewer 1	Accepted	No correction required
C003	Network Reviewer 2	Accepted	No correction required
C004	Network Reviewer 2	Accepted	No correction required
C005	Network Reviewer 3	Accepted	No correction required

Table 4.4: Responsible AI review summary

The five edited cases show why human review remains important even when the high-level diagnosis is useful. For example, C006 required verification of the ACL direction and HTTP-specific behavior; C015 required checking the OSPF network statement and area configuration; C020 required inspection of NAT rules and inside/outside interface definitions; C025 required verification of ACL direction and HTTPS requirements; and C030 required confirmation that the ACL was applied to the management interface in the correct direction..

4.9.6 Problems Faced and Solutions

Challenge	Solution
Connecting symptoms to root causes	Created evidence-to-diagnosis rules and prioritized specific rules.
Handling many Cisco fault types	Prepared a structured 30-case dataset across multiple networking concepts.
Preventing unsafe automatic fixes	Added mandatory human review and a Responsible AI log.
Checking diagnosis quality	Used deterministic checks and comparison with expected faults.
Presenting results clearly	Built separate Dashboard, Diagnose, and Responsible AI sections.

4.9.7 Skills and Learning Outcomes

- Cisco networking and troubleshooting commands.
- OSI model and fault isolation.
- Python programming and rule-based expert-system design.
- Structured prompt design and AI-assisted reasoning.
- CSV data preparation and evaluation.
- HTML/CSS/JavaScript dashboard development.
- Responsible AI and human-in-the-loop decision making.
- Team collaboration, testing, debugging, and technical documentation.

4.9.8 Limitations and Future Scope

The current system is designed for structured Cisco laboratory evidence and uses manually defined rules. It does not connect directly to live devices and does not automatically learn new rules. Its evaluation is based on a fixed dataset, so the 100% match result should be interpreted only as performance on the supplied test cases.

- Integrate an LLM for semantic understanding of free-form CLI output.
- Add secure live SSH, NETCONF, or RESTCONF integration.
- Expand the dataset to advanced routing, security, SD-WAN, and multi-vendor scenarios.
- Add an NLP/ML classifier for broader fault recognition.
- Add user authentication and collaborative case management.

- Integrate ticketing systems for operational workflows.

4.9.9 Conclusion

NetSage AI demonstrates how AI-assisted rule-based reasoning can support network troubleshooting in an academic Cisco lab environment. The system combines structured cases, evidence-based diagnosis, deterministic validation, a dashboard, and mandatory human review. The final implementation contains 30 troubleshooting cases and a Responsible AI workflow.

The project provided practical experience in networking, Artificial Intelligence, Python, data organization, web interface development, testing, and responsible technology use. The two-member approach divided the work into complementary technical areas while producing one integrated solution.

4.10 Certificates of Completion and Other Communication Mail Proof



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To: utkarshkumar5005@gmail.com

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